

Assembly changes what Vaxrank can claim

Actual RNA reconstruction, translation, epitope prediction, and vaccine-selection outcomes | decision matrix 1 of 3

A DNA event is not an antigen. Vaxrank requires a defensible mutant protein before ranking epitopes.

VARIANT	RNA / TRANSLATION EVIDENCE	VAXRANK OUTCOME
PIP5K1A p.G474fs Assembly-dependent frameshift indel	ONT assembly: 3 ALT, 0 OTHER; 49-aa mutant protein	SELECTED
TECPR1 p.Thr259fs Assembly-extended frameshift indel	Illumina assembly: 49 aa versus 37 aa without overlap assembly	SELECTED
GLIS3 p.Ser775fs Expressed 8-bp frameshift deletion	Illumina: 10 ALT, 1 OTHER; validated 43-aa frameshift context	SELECTED
RNF213 p.Ile1070Ile (del Leu) Expressed in-frame deletion	Illumina: 39 ALT observations / 38 templates; validated 49-aa context	SELECTED
DYNC1H1 p.Val314Ile RNA-supported missense control	ONT: 48 ALT, 7 OTHER; assembly and no-assembly sequences agree	SELECTED
DYNC1H1 p.Gln3267His Second RNA-supported DYNC1H1 locus	Illumina: 34 ALT, 0 OTHER; assembly adds one upstream residue	SELECTED
NAV2 p.Ala1809Val Multi-transcript RNA-supported missense	Illumina: 18 ALT; one sequence, two compatible transcript models	SELECTED

PREDICTION CONTEXT

NetMHCpan 4.2c; patient class-I alleles: HLA-A*01:01, HLA-B*08:01, HLA-B*27:05, HLA-C*01:02, HLA-C*07:01

Exploratory binding ranks: full-proteome exact-self screening was not run for explicit assembled-antigen inputs.

Assembly changes what Vaxrank can claim

Actual RNA reconstruction, translation, epitope prediction, and vaccine-selection outcomes | decision matrix 2 of 3

A DNA event is not an antigen. Vaxrank requires a defensible mutant protein before ranking epitopes.		
VARIANT	RNA / TRANSLATION EVIDENCE	VAXRANK OUTCOME
MAP2 compound frameshift haplotype RNA-resolved compound frameshift	T1 Illumina: 4 exact Q20 compound reads; T2 Illumina and ONT corroborate	SELECTED
NTF3 compound p.Lys69Ser haplotype RNA correction of isolated SNV effect	Illumina and ONT reproduce AG>GT; single-edit Lys→Arg is rejected	SELECTED
CD109 long-read phased alleles Long-read-resolved compound protein	ONT links both alleles in 21 UMIs; Illumina links neither	NO TARGET BINDER
ZNF436 p.Ser207Cys long-read context Long-read-expanded antigen context	ONT 49 aa; Illumina assembly 30 aa; Illumina unassembled 29 aa	SELECTED
FOXO3::STRADA/CCDC47 Cross-platform structural variant / fusion	55 DNA + 31 RNA split reads across four timepoints; donor frame unresolved	NO TRANSLATION
PARD3B::CDKN2B-AS1/CDKN2B Recurrent somatic translocation	119 DNA + 48 RNA split reads; direct ONT junction includes 12 inserted nt	NO TRANSLATION
AMPH 4.3-kb internal deletion DNA-supported intragenic structural deletion	DNA-supported 4.3-kb deletion; no RNA split reads	NO TRANSLATION
PREDICTION CONTEXT NetMHCpan 4.2c; patient class-I alleles: HLA-A*01:01, HLA-B*08:01, HLA-B*27:05, HLA-C*01:02, HLA-C*07:01 Exploratory binding ranks: full-proteome exact-self screening was not run for explicit assembled-antigen inputs.		

Assembly changes what Vaxrank can claim

Actual RNA reconstruction, translation, epitope prediction, and vaccine-selection outcomes | decision matrix 3 of 3

A DNA event is not an antigen. Vaxrank requires a defensible mutant protein before ranking epitopes.		
VARIANT	RNA / TRANSLATION EVIDENCE	VAXRANK OUTCOME
chr21 unnamed long-read-rich junction Long-read-dominant expressed structural junction	29 ONT/PacBio + 3 short-read RNA witnesses; unnamed CDS/frame	NO TRANSLATION
H1-2 repetitive 15-nt deletion Large in-frame deletion	ONT assembly: 13 ALT, 2 OTHER; no passing target binder	NO TARGET BINDER
ATP5MG::KMT2A Structural variant / gene fusion	One coding hypothesis; five donor-CDS-unavailable alternatives	HELD OUT
TPST1::CRCP T2 Structural variant / gene fusion	22 long reads rescue expression; CDS/frame remains unresolved	NO TRANSLATION

PREDICTION CONTEXT
NetMHCpan 4.2c; patient class-I alleles: HLA-A*01:01, HLA-B*08:01, HLA-B*27:05, HLA-C*01:02, HLA-C*07:01 Exploratory binding ranks: full-proteome exact-self screening was not run for explicit assembled-antigen inputs.

PIP5K1A p.G474fs

chr1:g.151242179delG (PIP5K1A p.G474fs) | Osteosarcoma T1 | Assembly-dependent frameshift indel

SELECTED

Long-read RNA assembly establishes an expressed frameshift protein and Vaxrank selects a supported long peptide.

1 DNA-ONLY INTERPRETATION

The genomic deletion predicts a frameshift, but DNA alone cannot establish the expressed haplotype, mutant protein extent, or junction-spanning peptide sequence.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T1 ONT RNA window

ALT 3 | REF 36 | OTHER 0 | top protein 2

Vaxrank RNA gates: **PASS**

3 ASSEMBLED OR TRANSLATED PROTEIN

VFKKIPLKPSPSKKFRSGSSFRRRAAPVATPALLTSHRSLGNTRHK

4 VACCINE-CONSTRUCT DECISION

SSFRRRAAPVATPALLTSHRSLGNT

Rank score 14.674 | target epitope score 8.472

RNA support × target epitope score
This is the positive assembly-dependent result: RNA reconstruction changes an abstract frameshift call into a ranked vaccine candidate with multiple class-I binders.

Selection is computational and is not a clinical recommendation.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
RRAAPVATPAL	HLA-C*01:02	116.92	0.352	1.254
AAPVATPAL	HLA-C*01:02	3.09	0.007	0.998
AAPVATPALL	HLA-C*01:02	18.97	0.080	0.988
RAAPVATPAL	HLA-C*01:02	19.49	0.082	0.988
APVATPAL	HLA-C*01:02	77.44	0.254	0.944

TECPR1 p.Thr259fs

chr7:g.98241128delG (TECPR1 p.Thr259fs) | Osteosarcoma T2 | Assembly-extended frameshift indel

SELECTED

Illumina overlap assembly extends the supported protein from 37 to 49 aa and Vaxrank identifies a target-bearing 25-aa construct.

1 DNA-ONLY INTERPRETATION

The one-base deletion predicts a frameshift, but DNA alone does not establish how much expressed mutant reading frame is supported.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina short-read RNA | T2 SARC0277; source 8fda918141cedf55

ALT 19 | REF 20 | OTHER 0 | top protein 15

Vaxrank RNA gates: PASS

3 ASSEMBLED OR TRANSLATED PROTEIN

SSWSLLDTPGEVVQISCGPHDLLWAHSGRDRPWSGKESTGAIPKEVPGP

4 VACCINE-CONSTRUCT DECISION

GPHDLLWAHSGRDRPWSGKESTGAI

Rank score 0.252 | target epitope score 0.252

source-agnostic target epitope score

This is the strongest short-read assembly example: both translations validate, while assembly adds twelve supported upstream residues and changes the candidate-window pool.

The osteosarc.com page publishes a longer predicted context; this panel ranks only the 49 aa directly reconstructed from the selected RNA evidence.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
GRDRPWSGK	HLA-B*27:05	531.38	0.919	0.252
SGKESTGAI	HLA-C*01:02	1881.61	3.088	0.000

GLIS3 p.Ser775fs

chr9:g.3856149_3856156delTGATGTGG | Osteosarcoma T1 | Expressed 8-bp frameshift deletion

SELECTED

Illumina RNA reconstructs a validated 43-aa mutant context and Vaxrank selects a target-bearing long peptide.

1 DNA-ONLY INTERPRETATION

The 8-bp deletion predicts a new reading frame but cannot prove that the resulting transcript and mutant tail are expressed.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina bulk RNA | T1 BostonGene; source 1b66c15da594a3ef

ALT 10 | REF 19 | OTHER 1 | top protein 10

Vaxrank RNA gates: PASS

3 ASSEMBLED OR TRANSLATED PROTEIN

QLPPLTAVDAGAERFAPSAPSPQPPESSSSFFNTAKNTASLYP

4 VACCINE-CONSTRUCT DECISION

SAPSPQPPESSSSFFNTAKNTASLY

Rank score 3.455 | target epitope score 3.455
source-agnostic target epitope score
RNA agrees with the nominated complex deletion, making GLIS3 a positive reconstruction control rather than a disagreement with Varcode.

The website's 80-aa vaccine sequence extends beyond the 43-aa RNA context ranked here and is retained separately in the source record.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
NTAKNTASL	HLA-C*01:02	55.54	0.190	0.962
FNTAKNTASL	HLA-C*01:02	174.46	0.477	0.837
NTAKNTASLY	HLA-A*01:01	187.43	0.058	0.820
FFNTAKNTASL	HLA-C*01:02	244.70	0.627	0.733
TAKNTASL	HLA-C*01:02	721.79	1.500	0.085

RNF213 p.Ile1070Ile (del Leu)

chr17:g.80327830_80327832delTAC | Osteosarcoma T1 | Expressed in-frame deletion

SELECTED

Illumina RNA supports the in-frame deletion and Vaxrank selects a target-bearing 25-aa construct.

1 DNA-ONLY INTERPRETATION

The genomic allele predicts loss of one leucine; RNA is needed to establish expression and its local translated context.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina bulk RNA | T1 BostonGene; source 1b66c15da594a3ef

ALT 39 | REF 17 | OTHER 16 | top protein 38

Vaxrank RNA gates: PASS

3 ASSEMBLED OR TRANSLATED PROTEIN

DILKHLTLADVKHVFRLCGTDEKIANVTEDAKRLIAVADSVLTKVVGD

4 VACCINE-CONSTRUCT DECISION

LTLADVKHVFRLCGTDEKIANVTED

Rank score 0.066 | target epitope score 0.066

source-agnostic target epitope score

The RNA and nominated edit agree, providing a robust complex-indel reconstruction control across the DNA, RNA, protein, and epitope layers.

The rank is specific to the five assessed patient class-I alleles and default filters.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
FRLCGTDEKI	HLA-B*27:05	761.12	1.173	0.066

DYNC1H1 p.Val314Ile

chr14:g.101980529G>A | Osteosarcoma T1 | RNA-supported missense control

SELECTED

RNA assembly and non-assembly modes agree on the same 45-aa context; Vaxrank recovers the strong KRFHATISF-centered candidate.

1 DNA-ONLY INTERPRETATION

The SNV predicts p.Val314Ile, but does not by itself establish expression or the sample-specific transcript context.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T1 ONT; source 53f498a544883d51

ALT 48 | REF 16 | OTHER 7 | top protein 39

Vaxrank RNA gates: **PASS**

3 ASSEMBLED OR TRANSLATED PROTEIN

RESPEVLLTLDILKHGKRFHATISFDTDGLKQALETVDYNPLM

4 VACCINE-CONSTRUCT DECISION

LTLTDILKHGKRFHATISFDTDGLK

Rank score 4.908 | target epitope score 4.908

source-agnostic target epitope score

This is a concordant control. Assembly does not change the sequence, and the selected window overlaps vaccine sequences published on [osteosarc.com](#).

RNA observations in the pinned primary corpus are evidence for reconstruction, not a full-sample abundance estimate.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
KRFHATISF	HLA-C*07:01	37.84	0.008	0.975
HGKRFHATI	HLA-B*08:01	160.43	0.201	0.855
GKRFHATISF	HLA-B*27:05	255.13	0.558	0.716
KRFHATISFD	HLA-B*27:05	291.76	0.618	0.654
ISFDTDGL	HLA-C*01:02	351.46	0.844	0.546

DYNC1H1 p.Gln3267His

chr14:g.102030200G>T | Osteosarcoma T1 | Second RNA-supported DYNC1H1 locus

SELECTED

Illumina assembly adds one supported upstream residue and Vaxrank retains a low-scoring but filter-passing target-bearing construct.

1 DNA-ONLY INTERPRETATION

The second DYNC1H1 SNV predicts p.Gln3267His but does not establish the expressed sequence context.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina bulk RNA | T0 BostonGene; source 1b66c15da594a3ef

ALT 34 | REF 17 | OTHER 0 | top protein 33

Vaxrank RNA gates: PASS

3 ASSEMBLED OR TRANSLATED PROTEIN

MVKDQQEAEKKKVMSQEIQEQLHKHQEVIADKQMSVKEDLDKVEPAVIE

4 VACCINE-CONSTRUCT DECISION

KKKVMSQEIQEQLHKHQEVIADKQM

Rank score 0.007 | target epitope score 0.007

source-agnostic target epitope score

This locus is distinct from the strong p.Val314Ile result. Its predicted class-I signal is weak, which is visible rather than hidden.

osteosarc.com annotates this locus but does not currently publish a protein sequence context; the displayed sequence is RNA-derived.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
EQLHKHQEV	HLA-B*08:01	1121.61	1.099	0.006
QLHKHQEVI	HLA-B*08:01	1516.53	1.456	0.000
QLHKHQEV	HLA-B*08:01	1533.19	1.470	0.000

NAV2 p.Ala1809Val

chr11:g.20080142C>T | Osteosarcoma T2 | Multi-transcript RNA-supported missense

SELECTED

Illumina RNA yields one validated 47-aa sequence compatible with two contributing transcript models and a target-bearing construct.

1 DNA-ONLY INTERPRETATION

The SNV maps to multiple transcript consequences; DNA alone cannot choose the RNA-contributing transcript model.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina short-read RNA | T2 SARC0277; source 8fda918141cedf55

ALT 18 | REF 16 | OTHER 0 | top protein 13

Vaxrank RNA gates: PASS

3 ASSEMBLED OR TRANSLATED PROTEIN

GTAQSADLRIRRQHSSDSVSSINSVTSHSSVGSNIESDSKKKKRKNW

4 VACCINE-CONSTRUCT DECISION

DLRIRRQHSSDSVSSINSVTSHSSV

Rank score 0.012 | target epitope score 0.012

source-agnostic target epitope score

The sequence is resolved while isoform identity remains non-unique. Both assembled transcript IDs are preserved instead of reporting one guessed source.

The RNA window ends before reference transcript predictions diverge, so this result does not claim unique isoform resolution.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
NSVTSHSSV	HLA-C*01:02	1029.88	1.951	0.012

MAP2 compound frameshift haplotype

chr2:g.209694769_209694800delinsAGGG | Osteosarcoma T1 | RNA-resolved compound frameshift

SELECTED

Four exact Q20 T1 Illumina reads establish the compound haplotype; its translation passes the evidence gate and is ranked as an explicit assembled antigen.

1 DNA-ONLY INTERPRETATION

The historical isolated 22-bp deletion predicts a different protein. Full evidence instead supports adjacent substitutions plus a 28-nt deletion on one haplotype.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina RNA with ONT corroboration | T1 UCLA; source 261f8be4ce5aa8c8

4 Q20 T1 ILMN + 2 Q20 T2 ILMN + 1 Q20 T2 ONT compound witnesses

Vaxrank RNA gates: PASS: EXPLICIT COMPOUND HAPLOTYPE

3 ASSEMBLED OR TRANSLATED PROTEIN

SQLEDRAHCHHLFKTVRIYQGRVVP

4 VACCINE-CONSTRUCT DECISION

SQLEDRAHCHHLFKTVRIYQGRVVP

Rank score 0.015 | target epitope score 0.015
source-agnostic target epitope score
The compound protein encodes ...DRAH..., not the listed-only ...DSTQ... or deletion-only ...DLAH... alternatives. The website now publishes the same compound protein context.

This is a transcript-aware reference-plus-compound-edit translation, not a whole-RNA Isovar protein reconstruction; that distinction is retained in the source metadata.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
VRIYQGRVV	HLA-B*27:05	1092.64	1.502	0.008
DRAHCHHLFK	HLA-B*27:05	1110.26	1.516	0.007
DRAHCHHLF	HLA-B*27:05	1903.18	2.160	0.000

NTF3 compound p.Lys69Ser haplotype

chr12:g.5494381_5494382delinsGT | Osteosarcoma T0/T1 | RNA correction of isolated SNV effect

SELECTED

The explicit compound allele is independently observed in Illumina and ONT RNA, passes the evidence gate, and yields a target-bearing peptide.

1 DNA-ONLY INTERPRETATION

Treating A>G alone predicts Lys-to-Arg. RNA carries the adjacent base change too, so the expressed AG>GT haplotype encodes serine.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina and Oxford Nanopore RNA | explicit compound stress corpus

4 ALT templates and 4 REF templates in the selected Illumina reconstruction

Vaxrank RNA gates: **PASS: EXPLICIT COMPOUND HAPLOTYPE**

3 ASSEMBLED OR TRANSLATED PROTEIN

NKLSKQMVDVSENYQSTLPK

4 VACCINE-CONSTRUCT DECISION

NKLSKQMVDVSENYQSTLPK

Rank score 0.842 | target epitope score 0.842

source-agnostic target epitope score

NTF3 is the cleanest RNA-versus-single-edit correction: the amino acid changes because the true local haplotype spans two adjacent bases.

osteosarc.com lists p.Lys69Ser but does not currently publish a protein context; the displayed window comes from the RNA reconstruction.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
VSENYQSTL	HLA-C*01:02	171.14	0.470	0.842

CD109 long-read phased alleles

chr6:g.73812224C>T phased with chr6:g.73818405G>T | Osteosarcoma T2 | Long-read-resolved compound protein

NO TARGET BINDER Twenty-one ONT UMI groups and three exact 222-nt Q20 reads phase the alleles, but no target epitope passes the configured class-I filters.

1 DNA-ONLY INTERPRETATION

Two variants 6,181 bp apart cannot be assigned to one expressed molecule from independent DNA calls or site-local short reads.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T2 ONT phased interval

21 ALT/ALT UMI groups; 3 exact full-interval Q20 reads; 0 linked Illumina reads

Vaxrank RNA gates: **PASS: LONG-READ PHASE**

3 ASSEMBLED OR TRANSLATED PROTEIN

QPMAVNISANGFGFAICQLNVVYNVKASGSSRRRRSIQNQEAFDLDAVAVKENKDDLNH
VDLNVCTSFSGPGMSG

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

Long reads change the protein claim by linking two substitutions 207 coding nucleotides apart. Vaxrank retains the negative epitope result without forcing selection.

No 25-mer spans both amino-acid changes; long reads prove molecular phase, while each local antigen remains independently targetable.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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No target epitope passed the configured filters.

ZNF436 p.Ser207Cys long-read context

chr1:g.23362762G>C | Osteosarcoma T1 | Long-read-expanded antigen context

SELECTED

ONT reconstructs 49 aa versus 30 aa with Illumina assembly and 29 aa without it, expanding the candidate pool and yielding a selected construct.

1 DNA-ONLY INTERPRETATION

The SNV predicts p.Ser207Cys but says nothing about how much contiguous expressed sequence is available for antigen-window enumeration.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore versus Illumina RNA I matched T1 comparison

ONT 3 ALT UMI groups; ILMN 8 ALT observations / 7 supporting templates

Vaxrank RNA gates: **PASS: LONG-READ CONTEXT**

3 ASSEMBLED OR TRANSLATED PROTEIN

QHQRTHTGERPYDCNECGKSFGRSCHLIQHQTIHTGEKPHKCNCEGKSF

4 VACCINE-CONSTRUCT DECISION

KSFGRSCHLIQHQTIHTGEKPHKCN

Rank score 0.517 | target epitope score 0.517
source-agnostic target epitope score
The mutation identity is unchanged; long reads expand the defensible flanking sequence and therefore the set of junction-overlapping long peptides Vaxrank can consider.

Three ALT UMI groups support the ONT context; platform-dependent context length is shown as evidence, not interpreted as differential abundance.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
KSFGRSCHL	HLA-C*01:02	444.04	0.996	0.382
GRSCHLIQH	HLA-B*27:05	645.26	1.050	0.134
GRSCHLIQHQ	HLA-B*27:05	1802.13	2.081	0.000
FGRSCHLI	HLA-B*08:01	2051.23	1.915	0.000

FOXO3::STRADA/CCDC47

chr6:108611245 ↔ chr17:63745980 | Osteosarcoma T1/T2 | Cross-platform structural variant / fusion

NO TRANSLATION

The top-ranked fusion has 55 DNA and 31 RNA split reads, yet the directly observed ONT junction window has no exact collinear annotated donor; translation is withheld.

1 DNA-ONLY INTERPRETATION

DNA split reads establish a recurrent somatic rearrangement, but breakends alone do not identify the expressed splice path, CDS start, or junction reading frame.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Short- and long-read RNA I fusion scan plus direct T2 ONT witnesses

55 tumor-DNA split reads; 31 RNA split reads; 3 direct T2 ONT junction fragments

Vaxrank RNA gates: **HELD OUT: NO ANNOTATED DONOR FRAME**

3 ASSEMBLED OR TRANSLATED PROTEIN

Not established

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

Strong cross-platform support establishes an expressed rearrangement, not an antigen. This is the clearest example of evidence strength and peptide readiness being separate questions.

A longer assembled transcript with justified CDS/frame is required before junction-spanning peptides can be enumerated.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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Not predicted because the observed junction has no justified donor CDS/frame.

PARD3B::

chr2:205310103 ↔ chr9:22007648 | Osteosarcoma T0-T3 | Recurrent somatic translocation

NO TRANSLATION

The positive-control fusion has 119 DNA and 48 RNA split reads; two direct T1 ONT witnesses include a 12-nt junction insertion, but no exact annotated donor CDS is established.

1 DNA-ONLY INTERPRETATION

The recurrent t(2;9) is a compelling somatic rearrangement but DNA does not specify the expressed transcript or reading frame.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Short- and long-read RNA I fusion scan plus direct T1 ONT witnesses

119 tumor-DNA split reads; 48 RNA split reads; 2 direct T1 ONT junction fragments

Vaxrank RNA gates: **HELD OUT: NO ANNOTATED DONOR FRAME**

3 ASSEMBLED OR TRANSLATED PROTEIN

Not established

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

The event is biologically credible and transcribed across timepoints, while its antigen sequence remains unresolved. Vaxrank therefore reports evidence but performs no peptide ranking.

The 12 inserted junction nucleotides and absent donor-frame assignment make naive reference-splicing translation unsafe.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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Not predicted because the sample-specific junction CDS/frame is unresolved.

AMPH 4.3-kb internal deletion

chr7:g.38543151_38547476del | Osteosarcoma T0 | DNA-supported intragenic structural deletion

NO TRANSLATION

No RNA split read supports the junction, so no sample-specific transcript, exon consequence, reading frame, or mutant protein is admitted.

1 DNA-ONLY INTERPRETATION

LINX/ESVEE resolves a 4.3-kb deletion with both breakends in AMPH and junction copy number 1.87.

2 SAMPLE-SPECIFIC RNA EVIDENCE

DNA WGS; RNA junction audit | T0 oncoanalyser SV

4.3-kb AMPH deletion; JCN 1.87; 0 RNA split reads

Vaxrank RNA gates: **HELD OUT: NO RNA JUNCTION**

3 ASSEMBLED OR TRANSLATED PROTEIN

Not established

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

This DNA-only event is a useful negative control: structural resolution at the genome level is insufficient for Vaxrank antigen construction.

Absence of RNA split reads may reflect absent expression or assay sensitivity; it is not proof that the DNA event is false.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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Not predicted because no expressed junction sequence is available.

chr21 unnamed long-read-rich junction

chr21:8397976 ↔ chr21:8442207 | Osteosarcoma T0-T3 | Long-read-dominant expressed structural junction

NO TRANSLATION

The junction has 32 RNA split reads, including 29 across ONT and PacBio, yet both partners are unnamed and no CDS start or frame is justified; translation is withheld.

1 DNA-ONLY INTERPRETATION

Two tumor-DNA split reads across T1 and T2 meet the site's multi-sample somatic threshold, but the breakends do not identify a coding transcript.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore, PacBio, and short-read RNA I fusion-wide 43-BAM junction audit

2 tumor-DNA split reads; 32 RNA split reads; 29 long-read witnesses

Vaxrank RNA gates: **HELD OUT: UNNAMED PARTNERS / NO CDS**

3 ASSEMBLED OR TRANSLATED PROTEIN

Not established

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

This is the clearest additional long-read-dependent expression example: long reads dominate the RNA evidence, but abundant support still cannot substitute for a defensible coding model.

The GC-rich chr21 locus has many nearby junction calls and unnamed partners, so mapping complexity and transcript identity must be resolved before antigen construction.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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Not predicted because neither partner has an identified coding transcript or reading frame.

H1-2 repetitive 15-nt deletion

chr6:g.26055825_26055839delCCTTGGGCTTAGCGG | Osteosarcoma T1 | Large in-frame deletion

NO TARGET BINDER

RNA assembly succeeds and passes Vaxrank's evidence gates, but no target epitope passes the default prediction filters.

1 DNA-ONLY INTERPRETATION

A 15-nt deletion inside a repetitive histone sequence does not by itself identify the expressed protein context or a useful class-I target.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T1 ONT RNA window

ALT 13 | REF 33 | OTHER 2 | top protein 11

Vaxrank RNA gates: **PASS**

3 ASSEMBLED OR TRANSLATED PROTEIN

VAKPKKAAKSAAKAVKPKVVKPKKAAPKKK

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

Assembly is necessary but not sufficient. The expressed sequence is resolved, yet Vaxrank correctly returns no vaccine candidate for the configured patient HLA set.

A zero-target result is retained rather than hidden or forced into selection.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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No target epitope passed the configured filters.

ATP5MG::

sample-specific fusion junction | Osteosarcoma T0 | Structural variant / gene fusion

HELD OUT

A coding fusion hypothesis exists, but competing noncoding donor hypotheses make the translated product ambiguous; Vaxrank holds it out.

1 DNA-ONLY INTERPRETATION

The DNA rearrangement supports a junction but cannot choose the sample-specific transcript, CDS start, reading frame, or translated junction product.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina short-read RNA assembly | T0 Personalis junction window

Junction assembly evaluated; allele-fragment counts are not available

Vaxrank RNA gates: **HELD OUT BEFORE RANKING**

3 ASSEMBLED OR TRANSLATED PROTEIN

MAQFVRNLVEKTPALVNG

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

The adapter preserves the plausible 18-aa translation for audit while preventing an ambiguous fusion from entering vaccine ranking.

Default NetMHCpan filtering also produced no targetable epitope for the coding hypothesis.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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No target epitope passed the configured filters.

TPST1::CRCP T2

DNA-positive, matched short-read-call-negative junction | Osteosarcoma T2 | Structural variant / gene fusion

NO TRANSLATION

DNA and 22 long reads support the junction, but the donor joins before its annotated CDS; no defensible fusion protein exists.

1 DNA-ONLY INTERPRETATION

The rearrangement is real enough to investigate, but a breakpoint is not a coding transcript and does not define a junction-spanning antigen.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T2 CTAT-LR-Fusion

22 junction-spanning long reads; no translated protein

Vaxrank RNA gates: **NOT APPLICABLE: NO PROTEIN**

3 ASSEMBLED OR TRANSLATED PROTEIN

Not established

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

This is a genuine long-read rescue of expression, not a Vaxrank antigen.
The protein and peptide claims remain withheld until the coding frame is established.

No translated sequence means no epitope prediction or vaccine ranking is scientifically valid.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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Not predicted because no translated sequence exists.

What is measured, inferred, and withheld

1 RECONSTRUCT

Isovar groups RNA fragments by allele, assembles supported sequence, and translates only when the coding frame is defensible.

EVIDENCE RETAINED

ALT, REF, and OTHER fragment counts; supporting-fragment count; platform; sample; transcript or junction provenance.

2 PREDICT AND RANK

Vaxrank enumerates mutant-protein windows and predicts class-I binding through mhctools and Topiary. Explicit assembled-antigen panels use target epitope score and do not claim a full-proteome exact-self screen.

ALLELE POLICY

The clinical null allele HLA-A*01:11N is not assessed. The five expressed class-I alleles are evaluated.

3 WITHHOLD WHEN NEEDED

Junction support alone does not establish a CDS, reading frame, translated product, or junction-spanning peptide. Ambiguous and unresolved fusion hypotheses remain outside vaccine selection.

NO EMPTY-CATEGORY FUDGING

Assembly success may still produce no target binder. That is a negative Vaxrank result, not a missing result.

SOFTWARE AND MODEL

isovar 1.17.0 | mhctools 3.44.0 | netMHCpan 4.2c | topiary 5.55.1 | vaxrank 3.19.0

<https://osteosarc.com/variant/PIP5K1A-chr1-151242178/>

[tests/data/osteosarc/selection_validation/documented.json](https://osteosarc.com/variant/TECPR1-chr7-98241127/)

<https://osteosarc.com/variant/TECPR1-chr7-98241127/>

[examples/osteosarc_complex_results/source/assembled_antigens.json](https://osteosarc.com/variant/TECPR1-chr7-98241127/)

<https://osteosarc.com/variant/GLIS3-chr9-3856149/>

<https://osteosarc.com/variant/RNF213-chr17-80327830/>

<https://osteosarc.com/variant/DYNC1H1-chr14-101980529/>

Exact evidence and scores are stored beside every figure in record.json.